

Mohamed Refaat

Mobile Software Engineer (Flutter)

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LinkedIn • GitHub • Portfolio

Summary

Mobile Software Engineer specializing in Flutter and Dart, with Excellent GPA (3.52/4.0) in Computing and Data Science. Experienced in architecting production-ready applications using Clean Architecture, BLoC/Cubit, and advanced backend integrations. Recently evolved from Firebase-based architectures to implementing offline first systems with PowerSync and Supabase, enabling real-time synchronization with automatic conflict resolution and zero data loss. Skilled in integrating AI-driven features (Google Gemini API), native Android services (AlarmManager, Background Tasks), and building responsive, adaptive UI/UX for mobile and desktop platforms. Committed to writing clean, testable code with comprehensive unit testing, optimizing performance, and delivering scalable, user-centric mobile solutions.

Education

Bachelor’s Degree in Computing and Data Science at Alexandria University, Alexandria, Egypt **October 2021 - July 2025**

Grade: Excellent (CGPA: 3.52 / 4.0)

Graduation Project: Egyptopia – AI-assisted tourism mobile application developed with Flutter, featuring personalized itinerary planning and real-time smart recommendations. (Grade: Excellent)

Experience

AI Systems Evaluator – Outlier (Remote) **June 2024 - Present**

- Evaluated and optimized AI chatbot responses based on accuracy, reasoning quality, and strict adherence to guidelines.
- Designed and tested complex prompt scenarios to assess AI model behavior across edge cases.
- Provided structured feedback to improve model consistency and response reliability.
- Worked within a distributed remote environment under strict quality standards and deadlines.

Mobile Development Intern – DEPI Program (MCIT), Skills Dynamix, Alexandria **October 2024 - June 2025**

- Participated in an intensive, hands-on training track focused on advanced Flutter and Android native development.
- Developed modular Flutter applications integrating RESTful APIs while applying UI/UX best practices and Clean Architecture.
- Applied Unit Testing fundamentals and technical documentation to ensure high-quality, maintainable codebases.
- Collaborated on team-based projects with structured codebases and feature-based development.

Technical Skills

Languages & Frameworks: Flutter, Dart, Java, Kotlin (Basics), Python, C++

Architecture & State Management: Clean Architecture, MVVM, SOLID Principles, BLoC/Cubit, GetX, GetIt (Dependency Injection), Feature Based Development

Backend & APIs: Firebase (Auth, Firestore, Storage), RESTful APIs, JSON, Django APIs, Supabase, PowerSync

Databases & Storage: • Cloud: Firebase Firestore, Supabase

• Local: Hive, SharedPreferences, GetStorage, Drift (SQLite), Local Database

Native & Advanced Features: Android AlarmManager, Local Notifications, AudioPlayers, Background Tasks, Camera & Image Picker Integration, Google Maps API & Location Services, Row Level Security (RLS)

AI & ML Integrations: Google Gemini API, AI-driven recommendations, Chatbot integration

Tools & Platforms: Git, GitHub, Postman, Figma, Google Cloud Console, Supabase Dashboard, PowerSync

Testing & Quality: Unit Testing, Debugging, Functional Testin, Performance Optimization

UI/UX & Visualization: FL Charts, Responsive Design (Mobile/Desktop adaptive), Custom Animations, Modular Widget Architecture, Dark/Light Theme Support

Projects

Egyptopia – Tourism Mobile App (Graduation Project) • [GitHub](#)

Flutter | BLoC & GetX | REST APIs | Django | Google Maps | Hive | AI Integrations Architected and developed a large-scale, comprehensive tourism platform using a hybrid state management approach (BLoC & GetX) adhering to Clean MVVM architecture.

Engineered an AI-Driven Itinerary Planner that dynamically generates personalized travel plans based on user budget, travel style, duration, and preferred categories via custom REST APIs.

Integrated a Smart Chatbot (Virtual Assistant) and an adaptive Recommendation Engine that tailors content to user preferences in real-time.

Implemented advanced features including real-time debounced global search, dynamic filtering (by city, rating, price), and a weather module with auto-location detection.

Enhanced user experience by integrating Google Maps deep linking for seamless navigation and utilizing Hive for high-performance offline storage of user wishlists.

Designed a secure role-based access system (Guest vs. Registered User) utilizing Firebase Authentication (Email & Google).

Slumber – Smart Sleep Tracking & Analytics App • [GitHub](#)

Flutter | BLoC | Firebase | FL Charts | AlarmManager | Local Notifications | AudioPlayers | Unit Testing

Architected a scalable, feature-modular sleep tracking application utilizing Clean Architecture, BLoC, and Firebase for robust state management and real-time data sync.

Developed a resilient background tracking system utilizing SharedPreferences and Android AlarmManager, ensuring zero data loss and triggering reliable Local Notifications (bedtime reminders & ongoing sessions) even when the app is completely terminated.

Engineered a custom algorithmic Sleep Score engine evaluating duration, consistency, and bedtime, visualizing complex health metrics (Sleep Debt, weekly trends) via interactive FL Charts.

Built a real-time Gamification & Achievements system powered by inter-Cubit communication to dynamically evaluate user habits and unlock milestones, enhancing user retention.

Integrated an ambient audio module using AudioPlayers, featuring seamless auto-looping and customizable sleep timers (15/30/60 mins) synchronized with sleep tracking.

Ensured software reliability by authoring 40+ comprehensive Unit Tests (100% passing) to validate core business logic, scoring algorithms, local data persistence, and midnight edge-case handling.

Memory Chat – Collaborative Note-Taking & Knowledge Management App • [GitHub](#)

Flutter | PowerSync | Supabase | Drift | Clean Architecture | BLoC | GetIt

Developed a production-grade offline-first mobile application with real-time synchronization across devices.

Implemented PowerSync + Supabase integration for seamless data synchronization with automatic conflict resolution.

Engineered a local-first architecture using Drift (SQLite) ensuring zero data loss and instant UI updates even without internet connectivity.

Built a hierarchical organization system (Workspaces → Sections → Memory Boxes → Notes) with full CRUD operations.

Designed a responsive Slack-inspired UI with adaptive layout (Drawer for mobile, persistent sidebar for desktop).

Implemented Row Level Security (RLS) policies and authentication flows for multi-user data isolation.

Plant Identifier – AI-Powered Plant Recognition App • [GitHub](#)

Flutter | Google Gemini API | Clean Architecture | Camera Integration

Developed an AI-driven plant identification app leveraging Google's Gemini API for image recognition.

Implemented Clean Architecture pattern for maintainable, testable codebase with clear separation of concerns.

Integrated camera and gallery functionality with platform-specific permissions (Android Manifest & iOS Info.plist).

Engineered a modular repository pattern enabling easy backend swapping without UI refactoring.

Languages

Arabic - Native

English - Proficient

French - Basic

Certifications & Training

Digital Egypt Pioneers Program (DEPI) – Mobile App Development from (MCIT)

Flutter Advanced Architecture (BLoC & MVVM) – Udemy

Flutter & Dart Fundamentals – Udemy